Object: The goal is to build an intelligent classification system that categorizes messy, free-text support tickets into specific departments (e.g., Billing, Technical Support, Account Recovery).

Install Libraries

```
!pip install groq pandas matplotlib seaborn scikit-learn
```

```
Collecting groq
  Downloading groq-1.2.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: typing-extensions<5,>=4.14 in /usr/local/lib/pyth
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.1
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
Downloading groq-1.2.0-py3-none-any.whl (142 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 142.3/142.3 kB 3.6 MB/s eta 0:00:00
Installing collected packages: groq
Successfully installed groq-1.2.0
```

Dataset Generation & Zero-Shot Classification

```
# !pip install groq pandas matplotlib seaborn scikit-learn  <-- Run this if lib

import pandas as pd
import os
import matplotlib.pyplot as plt
import seaborn as sns
from groq import Groq

# --- 1. INITIALIZE CLIENT ---
# Replace with your actual Groq API Key
client = Groq(api_key="PASTE_YOUR_GROQ_KEY_HERE")

# --- 2. CREATE SYNTHETIC DATASET ---
data = {
    'ticket_text': [
        "My account is locked and I can't reset the password.",
        "I was charged twice for my subscription this month.",
        "The mobile app keeps crashing whenever I try to upload a photo.",
        "I want to cancel my premium plan and get a refund.",
        "How do I integrate the API with my existing Python project?",
        "My laptop screen is flickering after the last software update.",
        "I didn't receive the OTP code on my registered mobile number.",
        "The website is loading very slowly in the Karachi region.",
        "Can I change my billing address from Karachi to Lahore?",
        "I forgot my username and the security questions are not working."
    ],
    'actual_category': [
        "Account Recovery", "Billing", "Technical Support", "Billing",
        "Technical Support", "Hardware", "Account Recovery",
        "Technical Support", "Billing", "Account Recovery"
    ]
}

df = pd.DataFrame(data)
```

Zero-Shot vs. Few-Shot Logic

```
# --- 3. CLASSIFICATION FUNCTIONS ---

def get_zero_shot_tags(text):
    """Classifies ticket with no prior examples."""
    prompt = f"""
    Categorize the following support ticket into the Top 3 most probable tags.
    Categories: [Billing, Technical Support, Account Recovery, Hardware, API/De

    Ticket: {text}

    Output format: 1. Tag1, 2. Tag2, 3. Tag3
    """
```

```
    response = client.chat.completions.create(
        messages=[{"role": "user", "content": prompt}],
        model="llama-3.1-8b-instant",
    )
    return response.choices[0].message.content

def get_few_shot_tags(text):
    """Classifies ticket using provided examples (Few-Shot)."""
    prompt = f"""
    You are a professional support ticket classifier.

    Examples:
    - "I can't log in to my profile" -> 1. Account Recovery, 2. Technical Suppc
    - "My credit card was declined" -> 1. Billing, 2. Payment Issue, 3. Account
    - "The API returns a 500 error" -> 1. API/Developer, 2. Technical Support,

    Now classify this:
    Ticket: {text}

    Output format: 1. Tag1, 2. Tag2, 3. Tag3
    """
    response = client.chat.completions.create(
        messages=[{"role": "user", "content": prompt}],
        model="llama-3.1-8b-instant",
    )
    return response.choices[0].message.content

# --- 4. EXECUTION ---
print("Processing Zero-Shot Classification...")
df['zero_shot_output'] = df['ticket_text'].apply(get_zero_shot_tags)

print("Processing Few-Shot Classification...")
df['few_shot_output'] = df['ticket_text'].apply(get_few_shot_tags)
```

```
Processing Zero-Shot Classification...
Processing Few-Shot Classification...
```

## Evaluation & Visualization

```
# --- 5. SIMPLE ACCURACY CALCULATION ---

# Extract the first tag from the LLM response strings
df['predicted_tag_zero'] = df['zero_shot_output'].str.extract(r'1\.\s*([^,]+)')
df['predicted_tag_few'] = df['few_shot_output'].str.extract(r'1\.\s*([^,]+)')

# Calculate Accuracy (Does the first tag match our actual category?)
zero_acc = (df['predicted_tag_zero'].str.strip() == df['actual_category']).mear
few_acc = (df['predicted_tag_few'].str.strip() == df['actual_category']).mean()

# --- 6. VISUALIZATION ---
```

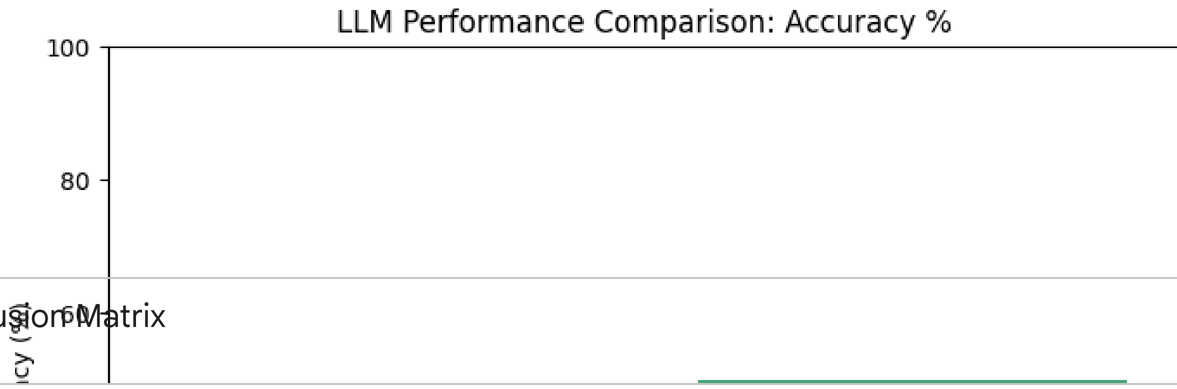
```
plt.figure(figsize=(8, 5))
sns.barplot(x=['Zero-Shot', 'Few-Shot'], y=[zero_acc, few_acc], palette='viridi
plt.title('LLM Performance Comparison: Accuracy %')
plt.ylabel('Accuracy (%)')
plt.ylim(0, 100)
plt.show()

# Display Final Table
print("\n--- FINAL CLASSIFICATION REPORT ---")
print(df[['ticket_text', 'actual_category', 'predicted_tag_zero', 'predicted_ta
```

```
/tmp/ipykernel_2655/3905764661.py:14: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in

  sns.barplot(x=['Zero-Shot', 'Few-Shot'], y=[zero_acc, few_acc], palette='vir
```




## Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

# Generate the report for Few-Shot (the better performer)
y_true = df['actual_category'].str.strip()
y_pred = df['predicted_tag_few'].str.strip()

# Print detailed metrics
print("\n--- Detailed Performance Metrics (Few-Shot) ---")
print(classification_report(y_true, y_pred))

# Plot Confusion Matrix
plt.figure(figsize=(10, 7))
cm = confusion_matrix(y_true, y_pred)
sns.heatmap(cm, annot=True, fmt='d', xticklabels=df['actual_category'].unique()
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix: Ticket Classification')
plt.show()
```

```
8  Can I change my billing address from Karachi t...            Billing
9  I forgot my username and the security question...   Account Recovery

                                  predicted_tag_zero  \
0  Account Recovery\n2. Technical Support\n3. Bil...
1  Billing\n2. Technical Support\n3. Account Reco...
2  Technical Support\n2. Hardware\n3. API/Develop...
3  Billing\n2. Account Recovery\n3. Technical Sup...
4  API/Developer\n2. Technical Support\n3. Billin...
5  Technical Support\n 2. Hardware\n 3. API/Devel...
6  Technical Support\n2. Account Recovery\n3. Bil...
7  Technical Support\n2. API/Developer\n3. Hardwa...
8  Billing\n2. Account Recovery\n3. Technical Sup...
9  Account Recovery\n2. Technical Support\n3. Bil...

                                   predicted_tag_few
0                                   Account Recovery
1                                            Billing
```

```
2                                              Mobile App
3                                                 Billing
4  API/Developer\n 2. Technical Support\n 3. Back...
5                                       Technical Support
6  Account Recovery\n2. Technical Support\n3. Sec...
7                                       Technical Support
8  Billing\n2. Account Information\n3. Profile Ma...
9                                        Account Recovery
```